

Caste, income, and the decision to write

A reproduction and interpretation of Dugar, Bhattacharya, and Reiley (2012)

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Abstract

Can higher income compensate for caste barriers in marriage search? We reproduce the principal tables in Dugar, Bhattacharya, and Reiley’s matrimonial-advertisement experiment and assess what its contacts reveal about this trade-off. Own caste receives the largest pooled share of letters within every respondent caste. That pattern does not distinguish own-caste attachment from a preference for rank or anticipated acceptance. A simple contact model shows how different caste valuations can generate identical letters when acceptance beliefs are unobserved. The reported income compensation therefore requires assumptions beyond reproduction of the response curve. It also depends on extrapolation and conceals opposite subgroup rankings: at Rs 35,000, the lower Brahmin attribute third sends 28 letters to the lower-caste groom and 8 to the Brahmin groom, while the upper third sends 9 and 76. The experiment documents caste sorting and income responsiveness in first contacts; it does not separately identify a monetary valuation of caste status.

1 Introduction

Can economic gains overcome caste barriers in marriage, and how much income would compensate for crossing those barriers? [Dugar et al. \[2012\]](#) address these questions with fictitious matrimonial advertisements that vary groom caste and income. The concern motivating the experiment is that caste prejudice excludes otherwise desirable partners. Its substantive claim is that higher income can offset that exclusion: the abstract describes a reduction in discriminatory behavior, and the introduction interprets the responses as substitution between caste status and groom income. Quantifying that substitution would help distinguish an absolute barrier from one that economic incentives can overcome.

The principal response counts, regressions, and compensation calculations reproduce. The interpretive difficulty is that a letter initiates a search rather than chooses a spouse from attainable alternatives. A family may write because it values a groom, expects acceptance, or has few attractive alternatives. Caste and income can affect each of those considerations. The experiment observes their combined consequence for contact, without separating a family’s caste valuation from its expectations about the other side.

Three findings organize this note. First, own caste is the largest pooled destination for each respondent caste. In the Brahmin own-versus-LC comparison, attachment to own caste and attraction to higher rank both favor the Brahmin groom. The other respondent castes make that distinction substantively relevant, although their contact patterns also reflect anticipated acceptance. Second, a simple model demonstrates that different caste valuations and acceptance beliefs can produce identical contact decisions. This limits the preference interpretation of the authors’ monetary estimates even if their contact effects are causal. Third, the Table 6 calculation depends on extrapolation and aggregation. Different Brahmin attribute groups rank the same high-income ads in opposite order.

The distinction between own-caste preference and rank matters for the economic consequences of caste.

In the same newspaper market, Banerjee et al. [2013] estimate strong own-caste preferences and use a matching model to show how those preferences can coexist with small equilibrium costs when suitable partners are available within caste. Their evidence concerns an earlier sample and includes evaluation of incoming letters, which already signal interest. It does not identify the mechanism behind the later experiment’s outgoing contacts. It does establish why a preference trade-off and a cost realized in equilibrium are different questions. We use that distinction to assess the authors’ claims, without treating any particular search mechanism as established by the letters.¹

2 The experiment, its outcomes, and its claims

2.1 Advertisements, respondents, and the observed decision

The authors placed nine fictitious brides-wanted advertisements in *Anandabazar Patrika*, a Bengali-language newspaper. The profiles cross three caste categories with three monthly incomes: Rs 7,000, Rs 15,000, and Rs 35,000. We denote these income levels LI, MI, and HI, respectively. Each profile appeared in two editions, producing eighteen placements. The data identify the editions as 2 and 16 September 2007. We use the paper’s codes HC, MC, and LC for the high-, middle-, and lower-caste categories, and R and G for respondent and groom. HC and MC denote Brahmin and Kayastha. The earlier draft identifies LC as Namasudra; the 2010 draft calls the category Scheduled Caste. We retain the category code when comparing versions rather than silently treating those labels as interchangeable.

All profiles describe government employment without specifying the job. The low-income profiles advertise a bachelor’s degree; the other profiles advertise a master’s degree. The low-to-middle comparison therefore changes education as well as income. Ages of 30–32 and heights of 64–66 inches were assigned within caste, according to the paper. There is one profile for each caste–income combination, repeated in the second edition, rather than several independently constructed profiles per combination [Dugar et al., 2012, author draft, section 3, pp. 10–16].

Interested families could send a letter to the advertisement’s post-office box. The experimental advertisements did not say “caste no bar.” They appeared among real advertisements in a newspaper that groups matrimonial notices by caste and sub-caste. The outcome is a first letter, before a meeting, acceptance, or marriage. We do not observe which columns a family read, which real advertisements it considered, or whether it contacted other advertisers. Posting the advertisements in the same edition makes them available; it does not establish that every family saw all of them.

2.2 Sample and outcomes

The advertisements received 1,366 letters. The data record the destination ad, respondent caste, newspaper edition, and reported bride and family attributes. Each letter identifies one contact; it does not establish that the family considered and rejected the other eight experimental ads.

Excluding the 243 flagged letters yields the paper’s 1,123-letter sample: 478 HC, 374 MC, and 271 LC responses. The paper describes excluding 47 non-unique letters from 22 respondents and letters from other caste groups. The exclusion indicator does not distinguish those reasons, and family identifiers are unavailable. We can reproduce the retained sample but cannot determine the reason for each excluded letter [Dugar et al., 2012, author draft, section 4 and footnote 12, pp. 16–17].

¹References to Dugar et al.’s sections, tables, figures, and pages use the December 2010 author version, <https://davidreiley.com/papers/CantBuyMeLove.pdf>. The published journal text has not been separately compared.

The authors also collected real advertisements to describe the market and inform the experimental profiles. That collection does not record each respondent’s available alternatives.

Table 4 counts letters. Table 5 fits a separate saturated linear probability model for each respondent caste, with eight groom indicators and LCG-LI as the reference category. Its fitted values equal shares of that caste’s retained letters sent to each ad. They sum to one across the nine ads. These are not response rates among all exposed families, because nonresponding readers are not observed. Original Figures 1–2 condition on groom income; original Figure 3 conditions on a respondent–groom caste pair. Those denominators must remain distinct when interpreting the plots.

2.3 From a contact share to an income compensation

Let $N_{c,g,y}$ be retained letters from respondent caste c to the groom of caste g with advertised income y . The fitted response probabilities in Table 5 are

$$\hat{p}_{c,g}(y) = \frac{N_{c,g,y}}{\sum_{g',y'} N_{c,g',y'}}.$$

They describe the destinations of retained contacts, conditional on a letter entering the sample. They are not the probability that an exposed family writes or that a marriage occurs.

The authors define discrimination behaviorally: fewer letters from higher-caste respondents to a lower-caste groom than to a comparable own-caste groom. More letters to richer lower-caste grooms constitute a reduction in that discriminatory behavior under their definition. A change in behavior need not mean a change in the underlying taste for caste. The stronger interpretive step is to read the response gradient as compensation for lost status [Dugar et al., 2012, pp. 5–6, 21–23].

Table 6 operationalizes compensation as the additional groom income predicted to equalize aggregate responses. With the own-caste groom’s income fixed at y_0 , the target is

$$p_{c,g}(y_0 + \Delta^{\text{contact}}) = p_{c,c}(y_0).$$

In the high-income Brahmin–LC comparison, the reported additional amount is Rs 35,950 on top of Rs 35,000. The authors describe such amounts as compensation to higher-caste respondents. An income that equalizes aggregate contact shares is not generally the mean or median income that makes individual families indifferent between attainable partners. Nor is groom income a cash payment to a respondent.

2.4 Numerical reproduction

All 27 response-count cells in original Table 4 reproduce exactly. Table 5’s coefficients, uncertainty entries, and fit statistics match to printed precision. Table 6 reproduces using the authors’ worksheet rules, including rounding and two significance-based zeroing choices in the Kayastha panel. Those rules change the reported monetary amounts without changing the underlying contacts. The appendix reports the reproduced exhibits, the unrestricted calculation, and the limited reporting discrepancies. The central issue is the interpretation of results that reproduce.

3 What would identify a valuation of caste?

3.1 Readership and the response baseline

Who reads the advertisements matters for the baseline against which contacts are compared. For one newspaper edition, let M_c be the number of eligible searching families of caste c , e_{cj} the fraction

exposed to ad j , and r_{cj} the average probability of writing among those exposed. Before sample exclusions, expected letters satisfy

$$\mathbb{E}[L_{cj}] = M_c e_{cj} r_{cj}.$$

For a fixed groom, the caste composition of received letters depends on the relative M_c as well as exposure and response behavior. A high Brahmin share of its letters cannot be interpreted as overrepresentation without a benchmark for eligible, exposed families. The Brahmin share of all newspaper readers would only approximate that population.

Within a respondent caste and edition, M_c cancels in the ratio of expected counts across two ads. Equal exposure would also cancel e_{cj} . Thus the overall Brahmin readership share is unnecessary for that relative comparison under common exposure. But caste-specific reading of columns can change exposure across ads; conditional response still includes attention, acceptance beliefs, and alternatives. The retained-letter analysis additionally requires attention to the exclusion rule. These are problems in interpreting this market's response baseline, independently of whether it represents a wider population.

A one-third own-caste benchmark does not follow from the experimental menu alone. It requires exchangeability of exposure and contact opportunities across the three groom castes at each income, including acceptance prospects and other profile attributes. Having three own-caste profiles and six other-caste profiles establishes availability in the newspaper, not those conditions.

3.2 A contact decision and an observational equivalence

Consider a family i and groom j . Let A_{ij} indicate that the family considers the ad, q_{ij} its perceived probability that contact leads to an attainable match, S_{ij} the value of that match relative to continuing search, and k_{ij} the cost of initiating contact. A simple decision rule is

$$D_{ij} = A_{ij} \mathbf{1}\{q_{ij} S_{ij} \geq k_{ij}\}. \quad (1)$$

The rule treats each contact as a marginal opportunity with a fixed continuation value. It permits several contacts and abstracts from interactions among simultaneous applications. It is an illustrative model, not an estimated description of these families.

Write match surplus as

$$S_{ij} = \beta_i y_j + \alpha_i \mathbf{1}\{g_j = c_i\} + \gamma_i s(g_j) + v_i(x_j) - O_i, \quad \beta_i > 0.$$

Here α_i values own caste, γ_i values a specified caste-rank score s , x_j contains other groom attributes, and O_i is the value of continuing search. Income is measured in thousands of rupees per month. A caste ordering does not supply a cardinal status scale, and these caste terms can represent perceived compatibility or social sanctions as well as the respondent's prejudice. For a Brahmin family, the additional LC groom income that holds match value constant, with other attributes fixed, is

$$\Delta_i^{\text{match}} = \frac{\alpha_i + \gamma_i \{s(HC) - s(LC)\}}{\beta_i}.$$

This is an income equivalent in match utility. Its interpretation as cash willingness to pay would require a further link between groom earnings and the family's welfare.

Unobserved acceptance beliefs prevent recovery of this quantity without restrictions. To see the problem directly, take positive surpluses $S_{ij} > 0$ and any nonnegative increment d_{ij} . Define

$$S'_{ij} = S_{ij} + d_{ij}, \quad q'_{ij} = q_{ij} \frac{S_{ij}}{S_{ij} + d_{ij}}.$$

The transformed beliefs remain between zero and one and satisfy $q'_{ij}S'_{ij} = q_{ij}S_{ij}$. Holding consideration and costs fixed therefore gives exactly the same contact decisions. Starting with no caste terms, an increment $d_{ij} = a_i \mathbf{1}\{g_j = c_i\} + b_i s(g_j)$, with nonnegative coefficients and rank score, can introduce own-caste or rank valuations while leaving every contact unchanged through the corresponding beliefs. Match compensation changes; observed letters do not.

This construction establishes nonidentification in the illustrative model when beliefs are unrestricted. It does not show that the constructed beliefs are the families' actual beliefs, that every search model is unidentified, or that caste prejudice is absent. Restrictions on acceptance probabilities can rule out the equivalence. The experiment contains no acceptance outcomes or elicited beliefs with which to impose or assess them. Fear of another family's prejudice may itself be a caste barrier, but it does not identify the letter writer's own valuation of caste.

3.3 Which assumptions could permit estimation?

Preference compensation could be estimated under a specified model in which consideration is observed or credibly restricted, acceptance and contact costs do not vary with the attributes used to identify preferences (or their variation is separately identified), and the distribution of utility heterogeneity and contact shocks is specified. Income and caste must enter the same identified utility model. A coefficient ratio would then have a model-based interpretation; a ratio of saturated contact-share differences is not automatically that ratio. Different families writing to different ads does not, by itself, preclude estimating population preferences under such a model.

Missing nonrespondents also need not defeat every relative-preference model. Under a correctly specified homogeneous multinomial logit with full consideration, mutually exclusive choices, and independence of irrelevant alternatives, ratios of inside-option probabilities cancel the outside-option denominator. But these families may contact several advertisers, consider different ads, and face different acceptance prospects. Treating every unchosen experimental profile as an observed rejection assumes away those problems. A flexible fit to the nine shares cannot test those assumptions.

Observing consideration directly is not the only possible route. [Abaluck and Adams-Prassl \[2021\]](#) show that, within specified consideration models, asymmetries in how choices respond to changes in competing alternatives' attributes can identify consideration and preferences separately. The identifying information is in those cross-responses under the model's restrictions. Repeating the same experimental menu does not supply independent changes in rival ads' attributes with which to recover that information. Applying such a model here would require an argument for identification beyond fitting the observed shares.

Search costs include the time and effort of inquiry and subsequent negotiation, not just postage. A shorter search horizon changes the continuation value O_i ; the availability of suitable own-caste partners can change both O_i and acceptance prospects. Background conditions common across ads do not automatically explain a caste contrast. The relevant question is whether they alter relative contact incentives or the composition of who writes. Holding the market fixed can suffice for an experimental effect on contacts. Predicting the cost of caste in completed matches requires a model of both sides and the available partners. The authors' appeal to simultaneously available own-caste ads and caste differences in average income does not establish which alternatives each respondent saw or could attain [[Dugar et al., 2012](#), pp. 26–27]. The real-ad collection is from May–June; it does not reconstruct the September respondents' choice sets.

Related studies obtain leverage from additional observations and restrictions. Speed-dating data make encounters explicit [[Fisman et al., 2008](#)]. Browsing and reply histories in online dating permit tests

of strategic initiation; those tests need not find strategic behavior, so it cannot simply be presumed here [Hitsch et al., 2010b,a]. In this newspaper market, Banerjee et al. [2013] find more evidence of strategic behavior in answering ads than in evaluating incoming letters and compare shortlisting with elicited rankings [pp. 57–59]. Their matching analysis then distinguishes preference trade-offs from equilibrium costs.

The existing data permit checks of income-specific patterns, editions, exclusions, and respondent attributes. Those checks can challenge a simple account of contact; they cannot validate unobserved exposure or beliefs. Recording ad views would measure exposure, and following replies and eliciting beliefs would constrain the acceptance channel, although replies observed only after contact would still be selected. Randomized information about acceptance could supply additional variation if it changes beliefs without directly changing match value. A “caste no bar” message does not automatically satisfy that exclusion: it may also change the appeal of the groom’s family. Independently replicated profiles and variation in competing menus would help separate profile effects and substitution. These are specific requirements for stronger inference, not adjustments available by adding controls to the current letters.

4 What the response patterns establish

4.1 Own caste and higher rank

Table 1 holds respondent caste fixed and shows where its letters went. Own-caste grooms receive the largest share within each respondent caste when incomes are pooled. This establishes own-caste concentration in recorded contacts. A model with only a common positive taste for higher rank, comparable non-caste attributes, and equal contact opportunities would instead favor Brahmin grooms for every respondent caste. The pooled pattern challenges that simple account. It does not distinguish an own-caste taste from caste-specific consideration or anticipated acceptance.

Table 1: Destinations of retained letters within respondent caste and income.

Respondent	Income (Rs)	Letters	To Brahmin (%)	To Kayastha (%)	To LC (%)
Brahmin	All	478	58.8	24.3	16.9
Brahmin	7,000	112	76.8	18.8	4.5
Brahmin	15,000	139	61.2	20.9	18.0
Brahmin	35,000	227	48.5	29.1	22.5
Kayastha	All	374	26.7	42.2	31.0
Kayastha	7,000	95	41.1	30.5	28.4
Kayastha	15,000	111	30.6	41.4	27.9
Kayastha	35,000	168	16.1	49.4	34.5
LC	All	271	11.4	29.9	58.7
LC	7,000	98	21.4	32.7	45.9
LC	15,000	91	7.7	42.9	49.5
LC	35,000	82	3.7	12.2	84.1

Notes: Each row uses letters from the specified respondent caste and, except for “All,” to grooms at the specified income. Percentages sum to 100 within a row. The “All” rows pool incomes; each groom caste has three profiles repeated in two editions. Other profile attributes are not identical, and the lowest income also carries lower education.

The distinction between own caste and higher rank is especially important for Kayastha and LC respondents. For Brahmins, both favor the same groom caste. The pooled own-caste ranking does not hold at every income: at Rs 7,000, Kayastha respondents send 39 letters to the Brahmin ad and 29 to their own-caste ad. A claim that own caste always dominates would therefore also overstate the data.

The authors discuss lower-caste behavior explicitly. Footnote 13 acknowledges own-caste concentration and the possibility that anticipated rejection discourages upward contacts. It says that separating the status incentive from discouragement requires information about men’s preferences that the study lacks. Footnote 14 nevertheless interprets the greater response to poorer higher-caste grooms as willingness to forgo income for status [Dugar et al., 2012, pp. 18–19]. For example, LC respondents send 7 letters to the Brahmin ad at Rs 15,000 and 3 at Rs 35,000. Both profiles advertise a master’s degree.

Within that comparison groom caste is unchanged. Greater contact with the poorer groom does not demonstrate an exchange of income for higher status. It is also consistent with targeting a more attainable prospect. Such behavior can coexist with a desire to marry up, but neither motive is identified by the inverse income gradient. The same standard applies to Brahmin contacts: anticipated acceptance and consideration must be addressed before interpreting their response gaps as caste valuations. Neither a common rank preference nor a purely horizontal preference is separately established by the full contact matrix.

4.2 Income responsiveness and the meaning of discrimination

At Rs 35,000, the Brahmin ad receives 110 Brahmin letters and the LC ad 51; at Rs 7,000 the counts are 86 and 5. Holding respondent caste fixed removes its overall readership size as an explanation of these within-caste differences under common exposure. It does not remove differences in consideration or expected acceptance. The low-to-high comparison also changes education.

A descriptive income elasticity can summarize the recorded pattern. Between the two master’s-degree LC profiles, the log-change slope in Brahmin contacts is

$$\frac{\log\{N_{HC,LC,35}/N_{HC,LC,15}\}}{\log(35/15)} = 0.84.$$

This finite contrast is not automatically a causal elasticity for changing the income of an otherwise fixed ad, or an elasticity of demand for completed marriages. With valid assignment, a total effect on contacts would be a meaningful experimental outcome even if it operates through attention or beliefs. Here each caste–income cell contains one profile repeated in two editions; letter-level precision does not provide independent profile replication.

Even if this were an identified contact effect, a positive income response could occur with a fixed caste penalty in utility. That is consistent with the authors’ behavioral definition of reduced discrimination. It does not establish a weakening of caste tastes, and the mapping from contact to income compensation still requires the assumptions above.

4.3 How many contacts cross caste?

Of the retained letters, 46.7 percent went to a groom from a different caste and 53.3 percent to an own-caste groom. As shares of all retained letters, 27.9 percent went to a lower-caste groom and 18.9 percent to a higher-caste groom. These are letters expressing initial interest, not completed marriages or the fraction of newspaper readers willing to marry outside caste.

Table 2: Share of letters from out-of-caste respondents, pooling incomes and editions.

Groom caste	Out-of-caste letters (%)	Letters received
Brahmin	31.8	412
Kayastha	55.5	355
LC	55.3	356
All	46.7	1123

Notes: The original retained sample is used. Out-of-caste means that respondent and groom caste differ. Each groom caste has three income profiles, each placed in two editions. Each percentage uses letters received by that groom caste as its denominator; the overall percentage uses all 1,123 retained letters. Including all 1,366 recorded letters gives an out-of-caste share of 51.0 percent; the exclusion reasons remain unresolved.

These percentages describe the composition of letters received. A groom caste’s own-caste share depends on the relative numbers of potential respondents from each caste as well as their contact behavior. The within-respondent comparisons above use a different denominator.

4.4 How much do extrapolation and the response curve matter?

For Brahmin responses to the LC groom at the highest income, the unrestricted formula gives an additional Rs 35,900, implying a total income of approximately Rs 70,900. The experimental maximum is Rs 35,000. The letter-bootstrap interval for the additional amount is approximately Rs 17,300–67,900. It quantifies sampling variation conditional on the observed advertisements and the letter-level independence assumption; it does not include uncertainty from changing the response curve or from ad-specific effects.

Table 3: Response-curve sensitivity: HCR–LCG at high income, thousand rupees.

Calculation	Additional income
Low-to-high slope (unrestricted Table 6)	35.9
Low-to-middle slope	23.6
Middle-to-high slope	45.4
Share linear in log income	245.5

These are alternative extrapolations, not four independently observed prices. The first three divide the same observed gap by different segment slopes. The last fits share as a linear function of log income and solves for the income matching the observed own-caste share. The low-to-middle slope also spans a change in advertised education. The experiment establishes responses at the three tested incomes; it does not select the curve outside that support.

The exclusions matter too. Retaining all recorded letters changes this unrestricted additional-income estimate to approximately Rs 19,800. That sensitivity does not establish that the excluded letters belong in the main sample: the underlying exclusion reasons remain unresolved. Letter-clustered and multinomial inference checks do not overturn the principal response contrasts. Repeating the same profiles in two editions does not independently replicate each treatment with new profiles.

4.5 Aggregate and subgroup response rankings

4.5.1 Grouping respondents by reported attributes

The original section 4.4 predicts that respondents with stronger attributes require more compensation, and describes adjusted trade-off estimates as close to the unadjusted ones [Dugar et al., 2012, pp. 23–25]. Its underlying estimates are not reported. We examine whether aggregate responses conceal different rankings across respondent groups: which ad attracts more letters from a given group at each tested income?

We summarize ten reported bride and family attributes using their first principal component, with median imputation for missing values. We form thirds within respondent caste, retaining tied scores together. Brahmin group sizes are 162, 157, 159. The index summarizes reported attributes; it is not a validated measure of desirability. Education groups provide a comparison that does not require an index. Full construction and missingness details are in the appendix.

4.5.2 A reversal at an observed income

At Rs 35,000, the lower third of Brahmin respondents sends 28 letters to the LC ad and 8 to the Brahmin ad. In the upper third the counts are 9 and 76. The LC-minus-Brahmin difference is 12.3 percentage points of lower-third letters (bootstrap 95% interval 4.2 to 18.1), compared with -42.1 points in the upper third (-50.9 to -31.4). The middle and upper groups favor the Brahmin ad at every tested income.

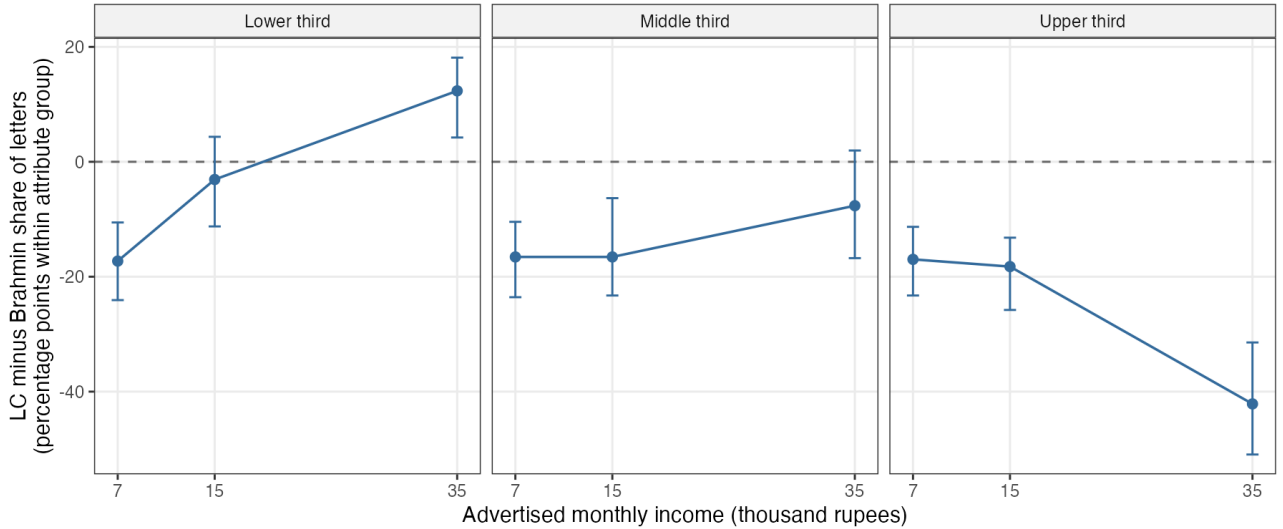


Figure 1: Brahmin respondents: LC minus Brahmin response shares within each attribute third. Each denominator includes all retained letters from that third across the nine ads. Positive values favor the LC ad; zero denotes equal counts. Points are observed at the three tested incomes; lines interpolate between them. Bars are pointwise 95% percentile intervals from 2,000 letter-level bootstrap draws within respondent caste, re-estimating imputation, the PCA and group boundaries. They condition on the observed profiles and assume independent letters. The low-to-middle comparison also changes groom education.

A direct test compares the distribution of Brahmin and LC destinations across the three attribute groups at each income. The Holm-adjusted Fisher p-values are 0.369, 0.000126, 4.81e-12, in ascending income order. These are conditional tests among letters to the two ads, with the estimated group assignments held fixed. They test subgroup differences, not whether one subgroup’s result is significant

and another’s is not. The bootstrap propagates uncertainty in the attribute index separately; neither procedure supplies randomization inference over new ads.

The lower-group reversal persists in both editions, when excluded letters are included, when education is omitted from the PCA, and among complete cases. Table 4 reports the high-income counts rather than a sequence of significance thresholds. Complete-case comparisons retain the original scores and boundaries and change the sample. They therefore address sensitivity to imputation, not selection into reporting all attributes.

Table 4: Letters at Rs 35,000 from lower and upper Brahmin attribute groups.

Sample	Lower: Brahmin	Lower: LC	Upper: Brahmin	Upper: LC
Retained	8	28	76	9
Edition 1	3	14	39	1
Edition 2	5	14	37	8
All recorded	10	39	83	11
Excluded only	2	11	7	2
PCA without education	14	24	69	9
Complete cases	5	14	48	7

Notes: Edition, exclusion, and complete-case checks use the retained sample’s scoring parameters and boundaries. The no-education specification re-estimates the index and boundaries on retained letters. Complete cases retain 260 of 478 Brahmin letters. Excluded letters are a sensitivity sample, not an established correction to the original exclusions.

Education alone gives the same ordering, with a small low-education group. At the highest income, Brahmin letters reporting less than a bachelor’s degree go 5 to the LC ad and 1 to the Brahmin ad. Letters reporting a master’s degree or doctorate go 13 and 68, respectively. These are selected groups of observed letter writers, not estimates of the causal effect of bride education on caste preferences.

4.5.3 The direction of crossing caste

The attribute gradient concerns the direction of contact as well as whether a letter crosses caste. Among Kayastha respondents, the share sent upward to Brahmin grooms rises from 10 percent in the lower third to 47 percent in the upper third. Among LC respondents, the upward share rises from 22 to 61 percent. The appendix reports all three destinations, including own-caste letters, for every group.

These patterns qualify an interpretation based only on horizontal own-caste preferences. They do not establish a vertical preference for higher caste: respondents with different attributes may also face different acceptance prospects, search costs, or alternatives. The design observes which advertised profiles attract which letters, not which feasible partners each family would choose.

4.6 A crossing within support answers a different question

Piecewise-linear interpolation puts the lower Brahmin third’s same-income crossing at Rs 19,000. There is no crossing within the tested range for the middle or upper third. The calculation moves both grooms’ incomes together until $p_{c,c}(y) = p_{c,LC}(y)$. Table 6 instead fixes the own-caste ad’s income and raises the other ad’s income. Neither comparison is an individual indifference point.

The crossing is less direct evidence than the observed high-income reversal. Of 2000 bootstrap draws, 3 have no lower-third crossing in the tested range. The corresponding numbers for the middle and upper thirds are 1844 and 2000. We retain those draws as outcomes rather than calculating an apparently precise unconditional interval from the subset that crosses. These frequencies describe the specified resampling exercise; they are not probabilities that a true crossing exists.

5 Implications

The response counts and central calculations reproduce. They establish caste sorting in first contacts and income responsiveness within particular respondent–groom pairs. Own caste is the largest pooled destination for every respondent caste, while respondent attributes substantially change the direction of cross-caste contact. These findings are compatible with several combinations of own-caste attachment, rank preference, and anticipated acceptance.

The Table 6 numbers answer an aggregate equal-contact question under an extrapolated response curve. Interpreting them as individual or average compensation for caste status requires additional identification. The contact model shows why observing the same letters can leave preference compensation different. The subgroup reversal adds an empirical limit: even the ranking of two ads at an observed income changes across respondents. This is consistent with heterogeneous responses; it does not supply an alternative WTP estimate.

These results support a claim about responses to advertised prospects in this marriage market. A monetary valuation of caste, a measure of respondents’ prejudice, and the cost of caste barriers in completed matches each require additional evidence and assumptions.

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Appendices

A Reproduction exhibits

A.1 Counts and regressions

All 27 response-count cells in Table 4 reproduce exactly. Table 5’s 54 coefficient and parenthesis entries and all 6 sample-size and fit entries match to printed precision. Table A1 reports the counts, and Table A2 reports the regression entries. The main comparison is visible without a model: at Rs 35,000, the Brahmin ad attracts 110 Brahmin letters and the lower-caste ad attracts 51. At Rs 7,000, the corresponding counts are 86 and 5.

Table A1: Reproduction of original Table 4: response letters.

Income	Groom	HCR	MCR	LCR
HI	HC	110	27	3
HI	MC	66	83	10
HI	LC	51	58	69
MI	HC	85	34	7
MI	MC	29	46	39
MI	LC	25	31	45
LI	HC	86	39	21
LI	MC	21	29	32
LI	LC	5	27	45

Table A2: Reproduction of original Table 5.

Groom	HCR	HC SE	MCR	MC SE	LCR	LC p
HCG-HI	0.2197	0.0198	0.0000	0.0190	-0.1550	0.000
HCG-MI	0.1674	0.0181	0.0187	0.0200	-0.1402	0.000
HCG-LI	0.1695	0.0182	0.0321	0.0207	-0.0886	0.002
MCG-HI	0.1276	0.0165	0.1497	0.0253	-0.1292	0.000
MCG-MI	0.0502	0.0119	0.0508	0.0217	-0.0221	0.477
MCG-LI	0.0335	0.0105	0.0053	0.0193	-0.0480	0.110
LCG-HI	0.0962	0.0149	0.0829	0.0230	0.0886	0.011
LCG-MI	0.0418	0.0112	0.0107	0.0196	0.0000	1.000
Constant	0.0105	0.0047	0.0722	0.0134	0.1661	0.000

Notes: HC and MC uncertainty entries are HC1 standard errors; the original LC column prints p-values despite its note calling them standard errors. Stacked observations are 4,302, 3,366, and 2,439; reported R^2 values are 0.05, 0.02, and 0.058. The omitted category is LCG-LI.

One reported test does not reproduce. Footnote 16 says the low-income Kayastha-versus-LC comparison is significant at the one-percent level; its reproduced p-value is 0.78. The body text correctly describes no significant difference. The other 20 footnote tests match the comparison criteria.

A.2 The Table 6 calculation

Let $p_{c,g}(y)$ denote the share of retained letters from respondent caste c sent to a groom of caste g at monthly income y , measured in thousands of rupees. The unrestricted calculation in footnote 21 is

$$\widehat{\Delta}_{c,g}(y) = \frac{\hat{p}_{c,c}(y) - \hat{p}_{c,g}(y)}{[\hat{p}_{c,g}(35) - \hat{p}_{c,g}(7)]/(35 - 7)}. \quad (\text{A1})$$

It converts a response-share gap into extra income by extending the lower-caste ad’s average low-to-high income slope.

Two Kayastha entries depend on rules documented in the authors’ calculation worksheet. It uses coefficients rounded to four decimals, treats the LCG-MI coefficient as zero because it is insignificant, and zeroes the low-income own-versus-LC gap for the same reason. Applying those rules reproduces all 15 printed Table 6 entries, including means and standard deviations. Table A3 displays the reconstruction alongside the unrestricted estimates.

Table A3: Reproduction of original Table 6: additional monthly income, thousand rupees.

Pair	Calculation	HI	MI	LI	Mean	SD
HCR-MCG	Worksheet	27.40	34.87	40.47	34.25	6.55
HCR-MCG	Unrestricted	27.38	34.84	40.44	34.22	6.56
HCR-MCG	Rounded unrestricted	27.40	34.87	40.47	34.25	6.55
HCR-LCG	Worksheet	35.95	36.56	49.33	40.61	7.56
HCR-LCG	Unrestricted	35.91	36.52	49.30	40.58	7.56
HCR-LCG	Rounded unrestricted	35.95	36.56	49.33	40.61	7.56
MCR-LCG	Worksheet	22.56	17.16	0.00	13.24	11.78
MCR-LCG	Unrestricted	22.58	13.55	1.81	12.65	10.42
MCR-LCG	Rounded unrestricted	22.56	13.54	1.79	12.63	10.42

Notes: Every worksheet entry rounds to its printed counterpart. “Unrestricted” uses full-precision coefficients and no zeroing; “Rounded unrestricted” uses four-decimal coefficients and no zeroing. The reported headline is an extra Rs 35,950 for an LC groom already advertising Rs 35,000.

A.3 Figures and their denominators

The following plots reproduce the original response shares. Original Figure 1 divides Brahmin letters by all Brahmin letters at each income. Original Figure 2 does the same for Kayastha respondents, including letters to the unplotted Brahmin grooms in the denominator. Its two displayed shares therefore do not sum to one. Original Figure 3 divides by all letters within each displayed respondent–groom caste pair, across incomes. The plotted values are tested directly against the counts in Table A1.

Original Figure 1. Brahmin responses by groom caste and income

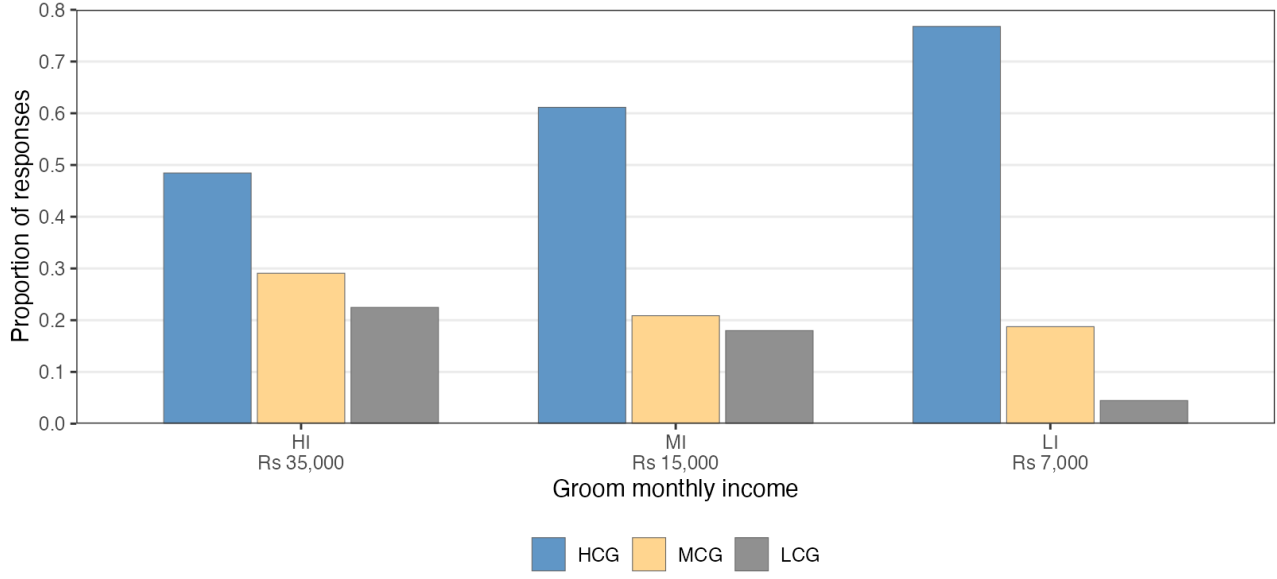


Figure A1: Reproduction of original Figure 1. Brahmin grooms receive the largest share of Brahmin letters at each income. HCG, MCG, and LCG denote the three groom categories; the denominator changes with income.

Original Figure 2. Kayastha responses to own-caste and lower-caste grooms

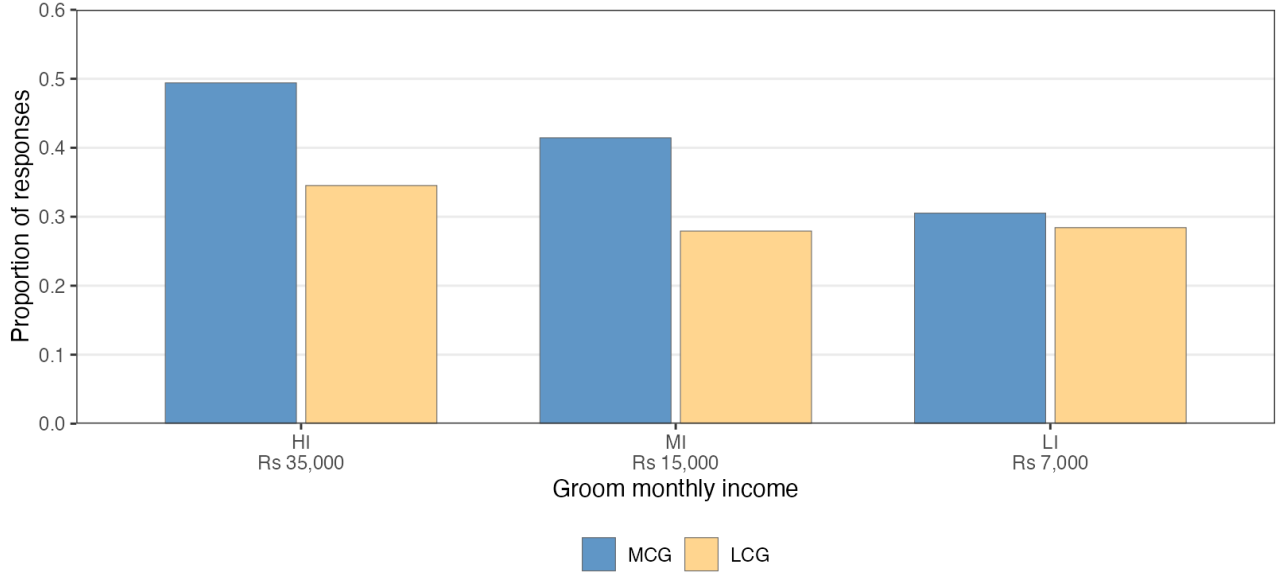
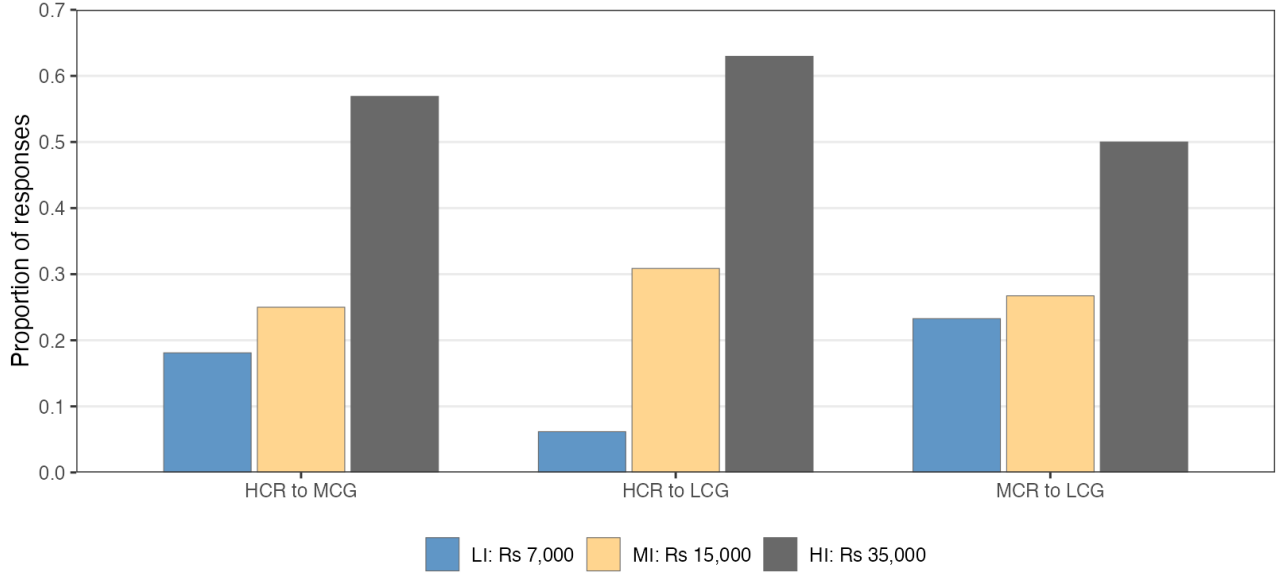


Figure A2: Reproduction of original Figure 2. Kayastha grooms receive more Kayastha letters than LC grooms, with almost no gap at low income. Letters to Brahmin grooms remain in each denominator.

Original Figure 3. Cross-caste responses by groom income



Denominator: letters within each pair (116; 81; 116), across all three incomes.
Source: Table 4; December 2010 draft, Figure 3.

Figure A3: Reproduction of original Figure 3. Higher-income grooms receive more letters in each displayed cross-caste pair. The three bars sum to one within a pair. HCR and MCR denote respondent categories.

B Attribute construction and supplementary comparisons

The PCA uses height, age, education, complexion, looks, employment, unmarried daughters, father absence, residence ownership, and car ownership. Ordinal attributes enter as numerical scores. Missing values receive the pooled retained-sample median for that attribute; variables are then centered and standardized. The first component is oriented by the summed loadings on education, complexion, looks, and residence ownership. Within-caste boundaries are empirical one-third and two-thirds quantiles, linearly interpolated between adjacent order statistics, with equality assigned to the lower group. Ties are not divided to force equal sizes.

Table B1: Index inputs, imputation, and missingness in retained letters.

Attribute	Imputation median	PC1 loading	Missing letters
Height (inches)	63	0.070	17
Age	25	0.203	15
Education	2	0.386	4
Complexion	2	0.343	93
Looks	1	0.403	158
Employed	0	0.357	0
Unmarried daughters	1	-0.333	0
Father absent	0	-0.198	0
Residence owned	0	0.369	408
Car owned	0	0.332	408

Residence and car reports are especially incomplete. The complete-case check keeps the original scoring rule and removes any letter missing one of its inputs. The no-education check instead fits a nine-attribute PCA on all retained letters, including re-estimation of its median imputations and boundaries. Neither exercise validates the index as an interpersonal scale of marital desirability.

An alternative split at the pooled index median also yields sharply different compensation estimates. For Brahmin responses to the LC groom at the highest income, the unrestricted Table 6 calculation gives Rs -14,000 in the lower half and Rs 129,500 in the upper half. These groups differ from the within-caste thirds used in the main analysis. Both comparisons describe heterogeneous response patterns; neither establishes group-specific willingness to pay.

Table B2: Letter destinations within respondent caste and attribute third.

Respondent	Third	Letters	Up (%)	Own (%)	Down (%)
HC	T1	162	0.0	35.8	64.2
HC	T2	157	0.0	58.0	42.0
HC	T3	159	0.0	83.0	17.0
MC	T1	125	10.4	52.0	37.6
MC	T2	124	22.6	45.2	32.3
MC	T3	125	47.2	29.6	23.2
LC	T1	92	21.7	78.3	0.0
LC	T2	89	41.6	58.4	0.0
LC	T3	90	61.1	38.9	0.0

Notes: Shares include all incomes and sum to 100 within a row. Upward destinations do not exist for HC respondents in this experimental menu, nor downward destinations for LC respondents; these zeros are structural.

Table B3: Brahmin letters by reported education and groom income.

Bride education	Group letters	Income (Rs)	Brahmin ad	LC ad
Below bachelor's	24	7,000	4	2
Below bachelor's	24	15,000	1	3
Below bachelor's	24	35,000	1	5
Bachelor's	267	7,000	50	3
Bachelor's	267	15,000	45	18
Bachelor's	267	35,000	41	33
Master's or doctorate	185	7,000	32	0
Master's or doctorate	185	15,000	38	3
Master's or doctorate	185	35,000	68	13
Missing education	2	7,000	0	0
Missing education	2	15,000	1	1
Missing education	2	35,000	0	0

The bootstrap resamples letters with replacement within respondent caste, preserving caste sample sizes. In each of 2,000 draws it re-estimates the medians, standardization, PCA, orientation, and

within-caste group boundaries. Draws with no crossing within the tested range remain in the analysis. Intervals are pointwise percentile intervals for the response-share differences. They reflect the assumed letter sampling process, including estimation of the index; they do not cover unidentified family dependence, selection into responding, or profile-level variation. The subgroup analyses are exploratory and use both editions; neither edition is a held-out confirmation sample.

C Replication scope

The replication covers the response counts, regressions, income-compensation calculations, and figures in original Tables 4–6 and Figures 1–3. The real-advertisement summaries in Table 2 do not fully reproduce from the available data. The unreported section 4.4 estimates cannot be compared cell by cell without the authors’ underlying results. Our subgroup analysis therefore evaluates the substantive heterogeneity claim rather than reproducing those unreported coefficients.